

CAPITAL UNIVERSITY

Alumni Portal

A Bridge Between the University and its Graduates

Faculty of Computers & Artificial Intelligence | Software Engineering Department

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Meet The Team

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Presentation Outline

1. **CHAPTER 1:** Introduction & Problem Statement
2. **CHAPTER 2:** Planning & Analysis
3. **CHAPTER 3:** Software Design

1. **CHAPTER 4:** Implementation Pipeline & Tech Stack
2. **CHAPTER 5:** System Workflows & Access Matrix
3. **CHAPTER 6:** Expected Impact & Conclusion

CHAPTER 1: INTRODUCTION

Project Overview & The Challenges

Project Overview: Transforming university-alumni relations through a unified digital networking and administrative ecosystem.

- **Communication Gap:** Graduates often lose touch with their alma mater and fellow peers shortly after graduation, leading to a loss of valuable professional networks.
- **Data Fragmentation:** Alumni records are often scattered across manual systems or disconnected social media groups, making it difficult for the university to track success metrics.
- **Request Delays:** Traditional methods for requesting official documents and certificates are slow, tedious, and require physical presence on campus.
- **Missed Opportunities:** Fresh graduates struggle to find credible career guidance and professional job openings specifically tailored to their faculty's expertise.

CHAPTER 1: INTRODUCTION

Key Objectives (Proposed Solution)

Networking

Foster a strong, verified graduate community for continuous knowledge sharing, professional collaboration, and structured mentorship.

Career Growth

An interactive job board linking graduates, students, and companies. Alumni and the university can directly publish and share job offers, internships, and events, building an integrated professional network within the campus community.

Digitization

Automate administrative tasks, enabling secure online document requests, modern automated approval flows, and unique digital graduation IDs.

System Requirements: Alumni Experience

Smart Profiles

Highlight graduation data, post-grad achievements, and professional skills in verified, structured profiles.

Social Feed

Exchange industry knowledge, post career milestones, like, comment, and communicate in real-time.

Dedicated Hubs

Join tailored professional channels organized by graduation year, specific specialization, or current industry.

Verified Digital ID

Leverage a unique, QR-enabled university-verified card acting as a modern physical graduation identity.

CHAPTER 2: PLANNING & ANALYSIS

Administrative Power & Governance

The Centralized Admin Dashboard delivers total governing visibility over the entire platform, ensuring data safety and request compliance.

- **Platform Analytics:** Analyze critical platform metrics, including total active users, verification statuses, posts, and dynamic system interaction levels.
- **Account Verification:** Validate graduates using an in-house developed **Central Alumni Database System**, serving as a unified integration gateway.
- **Document Approvals:** Authorize official university documents, certificate prints, and transcripts in a secure, highly organized workflow.

System Workflow Architecture

01

Registration

Graduates register online using their personal credentials, including National ID security checks.

02

Verification

Admin cross-checks details via the internally-developed Central Alumni Database System to link graduate records.

03

Interaction

Alumni gain full authorized access to communities, messaging feeds, and exclusive job opportunities.

04

Services

Users easily order verified paperwork, process clearance status, or export digital student profiles.

CHAPTER 4: IMPLEMENTATION PIPELINE

Technology Stack Overview

Frontend Hub: Powered by **React.js** for component-based, high-performance, and responsive user interfaces.

Backend Services: Built using **Node.js** and **Express** for high-throughput API routing and swift query orchestration.

Unified DB Layer: **PostgreSQL** combined with our custom Central Alumni Database System as a secure single point of truth.

Real-Time Systems: **WebSockets** protocol powering interactive messaging components and live notification threads.

CHAPTER 5: SYSTEM WORKFLOWS

Roles & Access Control Matrix

Ecosystem Features	Alumni (Graduates)	Staff / Department	Super Admin
Post Articles & Job Ads	✓	✓	✓
Request University Documentation	✓	✗	✗
Approve Requests & Verify Profiles	✗	✓	✓
Configure System Modules & DB Keys	✗	✗	✓

CHAPTER 6: EXPECTED IMPACT

Expected System Performance Metrics

100%

Automation

Transition to fully paperless digital verification and administrative workflows.

85%

Acceleration

Reduction in document processing and operational turnaround times.

3x

Engagement

Increase in active peer-to-peer professional communication and student support.

Unified

Central Database

Single connection gateway to bridge core cross-service platform dependencies.

Conclusion & Future Scope

The Alumni Portal serves as vital digital infrastructure for Capital University. It bridges the gap between graduating years and ongoing professional success while successfully modernizing administrative tasks through standard Software Engineering practices.

Future Scope:

- **AI-Driven Vacancy Matching:** Intelligent automation algorithms connecting graduates to tailored job postings based on profile data tracking.
- **Native Mobile Applications:** Deploying native iOS and Android mobile platforms to improve active on-the-go communications.

Thank You!

Any Questions?